This exam has 6 pages and 7 exercises.

The result will be computed as (total number of points plus 10) divided by 10.

Please motivate all answers!

1. **Equivalence relations** (15 points)
Suppose that in the set $\mathbb{Z}^2$ (the set of all pairs of integer numbers), we define the following relation:

$$\langle a, b\rangle \; R \; \langle c, d\rangle \qquad \iff \qquad \text{either } a \leq d \text{ or } b < c \quad (\text{this is an } \textit{exclusive} \text{ or})$$

where $\leq$ and $<$ are the usual inequalities between integer numbers.

Determine whether the relation R is reflexive, symmetric, and/or transitive, and explain your answer in each case. Conclude from your answers whether or not the relation R is an equivalence relation.

Solution: For any pair of integers $\langle a, b\rangle$ it is the case that either $a \leq b$ is true and $b < a$ is false, or else $a \leq b$ is false and $b < a$ is true. It follows that $\langle a, b\rangle \; R \; \langle a, b\rangle$ holds, so R is reflexive.

To check for symmetry, suppose that $\langle a, b\rangle \; R \; \langle c, d\rangle$ holds. There are two possibilities:

- $a \leq d$ and $b \geq c$. In this case, $c \leq b$ and $d \geq a$, and therefore $c \leq b$ is true while $d < a$ is false. Hence $\langle c, d\rangle \; R \; \langle a, b\rangle$.
- $a > d$ and $b < c$. In this case, $c > b$ and $d < a$, so $c \leq b$ is false while $d < a$ is true. It again follows that $\langle c, d\rangle \; R \; \langle a, b\rangle$.

We conclude that R is symmetric.

However, R is not transitive. For example, $\langle 1, 3\rangle \, R \, \langle 2, 2\rangle$ since $1 \leq 2$ is true and $3 < 2$ is false, and $\langle 2, 2\rangle \, R \, \langle 4, 1\rangle$ since $2 \leq 1$ is false and $2 < 4$ is true. But it is not the case that $\langle 1, 3\rangle \; R \; \langle 4, 1\rangle$, because $1 \leq 1$ is true and $3 < 4$ is true as well.

Since R is not transitive, it is not an equivalence relation.

2. **Functions** (6 + 4 + 5 points)
Consider the functions $abs\colon \mathbb{R} \to \mathbb{R}$ and $add_3\colon \mathbb{R} \to \mathbb{R}$ defined by

$$abs(x) = |x| \qquad\qquad add_3(x) = x + 3$$

(a) For each function, determine whether or not it is injective and whether or not it is surjective. Please give a brief explanation of your answer in each case.

Solution: The function abs is not injective, since $abs(-1) = abs(1) = 1$. It is not surjective either, since there is no real number x such that $abs(x) = -1$.

The function add_3 is injective, since $add_3(x) = add_3(y)$, i.e. $x+3 = y+3$ (with x and y real numbers), implies that $x = y$. It is also surjective, since for every real number y, $add_3(x) = y$ will be true for $x = y - 3 \in \mathbb{R}$.

(b) Give an explicit description (i.e. a formula for $f(x)$) of the function f defined by

$$f := add_3 \circ abs.$$

Solution:

$$f(x) = |x| + 3$$

(c) Express the function $g\colon \mathbb{R} \to \mathbb{R}$ defined by the description

$$g(x) := |x - 6|$$

as a composition of the functions abs, add_3, and their inverses.

Solution:

$$g = abs \circ add_3^{-1} \circ add_3^{-1}$$

3. Induction (5 + 10 points)

We claim that the following statement is true for all integer numbers $n \geq 0$:

$$\sum_{k=0}^{n} 2(k+1) = (n+1)(n+2)$$

(recall that $\sum_{k=0}^{n} a_k$ is shorthand notation for $a_0 + a_1 + a_2 + \cdots + a_n$).

(a) Verify by explicit calculation that the statement is true for $n = 0$, $n = 1$ and $n = 2$.

Solution:

	$2(n+1)$	$\sum_{k=0}^{n} 2(k+1)$	$(n+1)(n+2)$
$n = 0$	$2 \cdot 1 = 2$	2	$1 \cdot 2 = 2$
$n = 1$	$2 \cdot 2 = 4$	$2 + 4 = 6$	$2 \cdot 3 = 6$
$n = 2$	$2 \cdot 3 = 6$	$2 + 4 + 6 = 12$	$3 \cdot 4 = 12$

So the statement is indeed true for $n = 0$, $n = 1$ and $n = 2$.

(b) Using mathematical induction, prove that the statement is true for *all* integer numbers $n \geq 0$.

Solution: We have already shown in part (a) that the statement is true for $n = 0$ (the base case). For the induction step, let $m \geq 0$ be arbitrary, and assume (as induction hypothesis) that

$$\sum_{k=0}^{m} 2(k+1) = (m+1)(m+2). \qquad \text{IH}$$

Then, on the one hand,

$$\begin{aligned}
\sum_{k=0}^{m+1} 2(k+1) &= \sum_{k=0}^{m} 2(k+1) + 2((m+1)+1) \\
&\overset{\text{IH}}{=} (m+1)(m+2) + 2(m+2) \\
&= ((m+1)+2)(m+2) \\
&= (m+2)(m+3).
\end{aligned}$$

On the other hand,

$$((m+1)+1)((m+1)+2) = (m+2)(m+3).$$

Therefore, the induction hypothesis for $m \geq 0$ implies that

$$\sum_{k=0}^{m+1} 2(k+1) = ((m+1)+1)((m+1)+2).$$

Together with the base case, this proves that the statement in the claim is true for every integer $n \geq 0$.

4. Equivalence classes for $\equiv$ (8 points)

Group the following ten formulas into their semantic equivalence classes:

$$\top \qquad \bot \qquad q \qquad \neg p \vee q \qquad \neg p \wedge q \qquad p \to q$$

$$p \to (p \to q) \qquad (p \to p) \to q \qquad p \to (q \to q) \qquad (p \to q) \to q$$

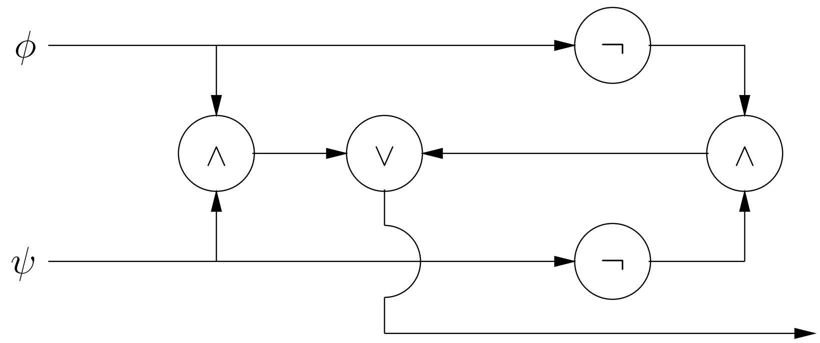
Solution:

- $\top \quad p \to (q \to q)$
- $q \quad (p \to p) \to q$
- $\neg p \vee q \quad p \to q \quad p \to (p \to q)$
- $\neg p \wedge q$
- $(p \to q) \to q$
- $\bot$

5. Logic circuits (12 points)

Give a logic circuit corresponding to the formula $\phi \leftrightarrow \psi$, using only AND, OR and NOT gates.

Solution: $\phi \leftrightarrow \psi$ is semantically equivalent to $(\phi \wedge \psi) \vee (\neg\phi \wedge \neg\psi)$. Hence a logic circuit for $\phi \leftrightarrow \psi$ is:



One alternative is to build a circuit that expresses $(\neg\phi \vee \psi) \wedge (\neg\psi \vee \phi)$.

6. OBDDs (9+4 points)

Let O_1 and O_2 be OBDDs, against the same variable order.

(a) Explain how an OBDD for $O_1 \to O_2$ can be constructed from O_1 and O_2.

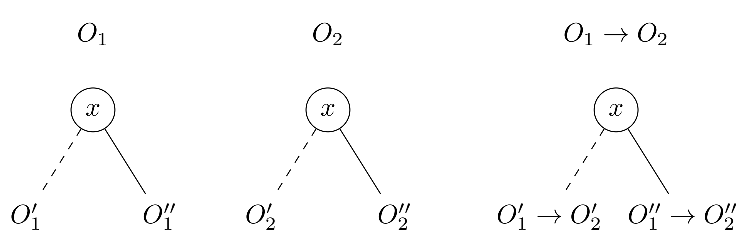
Solution: This is basically done in the same way as the construction of an OBDD for $O_1 \vee O_2$. The only difference is if O_1 or O_2 is a leaf.

Suppose O_1 is a leaf. If it is the leaf 0, then return the leaf 1. If on the other hand it is the leaf 1, then return O_2.

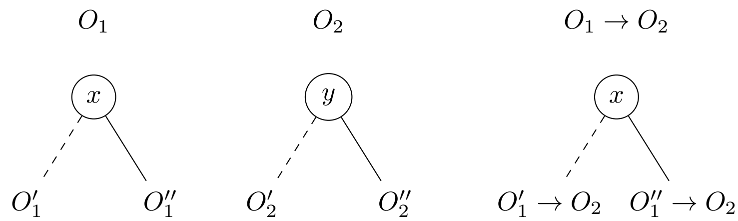
Likewise, suppose O_2 is a leaf. If it is the leaf 0, then return $\neg O_1$ (which is obtained by swapping 0 and 1 in the two leaf nodes of O_1). If on the other hand it is the leaf 1, then return the leaf 1.

Now suppose O_1 and O_2 are both not single leaves. As with disjunction, there are two cases.

Case 1: The roots of O_1 and O_2 are associated to the same variable x. Then the OBDD for $O_1 \to O_2$ is constructed as follows.



Case 2: The roots of O_1 and O_2 are associated to different variables x and y. If x is before y in the variable order, then the OBDD for $O_1 \to O_2$ is constructed as follows.



The construction is likewise if y is before x in the variable order, but then the root is y and recursively $O_1 \to O_2'$ and $O_1 \to O_2''$ need to be constructed.

An alternative solution is to rewrite $O_1 \to O_2$ into $\neg O_1 \vee O_2$ and use the standard operations $\neg$ and $\vee$ on OBDDs.

(b) Give an example to show that given reduced OBDDs O_1 and O_2, the constructed OBDD for $O_1 \to O_2$ is not always reduced.

Solution: Take for example the OBDDs for x and x. The constructed OBDD for $x \to x$ consists of root variable x with the 0- end 1-edge both pointing to a leaf node 1. While the reduced OBDD for $x \to x$, a tautology, consists of a single leaf node 1.

7. Predicate logic (3 + 3 + 6 points)
Consider the following two unary predicates:

- $C(x)$: x is a car
- $F(x)$: x has four wheels

(a) Express the phrase "Every car has four wheels" in predicate logic.

Also express the phrase "If all elements are cars, then all elements have four wheels" in predicate logic. ("All elements" refers to the set A of the model.)

Solution: $\forall x(C(x) \to F(x))$

$\forall x(C(x)) \to \forall x(F(x))$

(b) Explain why the first phrase semantically entails the second phrase.

Solution: Consider any model $\mathcal{M}$ in which $\forall x(C(x) \to F(x))$ holds. Then $C^{\mathcal{M}} \subseteq F^{\mathcal{M}}$, meaning that in the set A of elements associated to $\mathcal{M}$, for each element for which C holds, also F holds.

Hence, if C holds for all elements in A, then F holds for all elements in A. This implies that $\forall x(C(x)) \to \forall x(F(x))$ holds in $\mathcal{M}$.

(c) Give a model to show that the second phrase does not semantically entail the first phrase.

Solution: Consider a model $\mathcal{M}$ with $A = \{a, b\}$. Let C hold for a bit not for b, and let F not hold for a.

$\forall x(C(x)) \to \forall x(F(x))$ holds in $\mathcal{M}$, because $\forall x(C(x))$ is false, since C does not hold for b.

But $\forall x(C(x) \to F(x))$ does not hold in $\mathcal{M}$, because C holds for a but F does not.